WORKED SOLUTION

STEP I 1993, Question 11

Mechanics • Centre of Mass

STEP I 1993, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q11

The key idea

A rigid body hanging freely from one point is acted on by just two forces: the pull of the support and its weight. Two forces can balance only if they act along the same line, so the centre of mass must hang directly below the point of support. Everything therefore comes down to finding the centre of mass G of the bent wire, and then noticing that the body turns, between the two positions, through the angle between the lines GA and GB.

Solution

The centre of mass

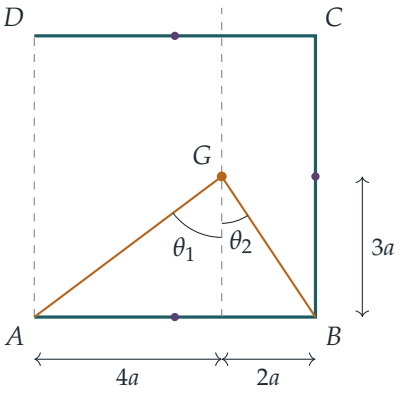
Model the wire as a uniform rigid rod bent into the three sides AB, BC, CD, each of the same weight W, hung from a smooth peg or a light string. To keep the numbers whole, take the side of the square to be $6a$ and set up axes in the plane of the wire with

$$A = (0,0), \quad B = (6a,0), \quad C = (6a,6a), \quad D = (0,6a).$$

Each side is a uniform rod, so its weight acts at its midpoint: $(3a,0)$ for AB, $(6a,3a)$ for BC and $(3a,6a)$ for CD. The three weights are equal, so G is the average of the three midpoints:

$$G = \left(\frac{3a+6a+3a}{3}, \frac{0+3a+6a}{3}\right) = (4a,\, 3a).$$

(By symmetry about the line through the midpoints of BC and AD, G had to lie at height $3a$; only its horizontal position needed working out.)



The dashed vertical line through G in the figure is parallel to BC (and to the missing side AD). Let θ_1 be the angle GA makes with it and θ_2 the angle GB makes with it, on the other side. Reading off the figure,

$$\tan\theta_1 = \frac{4a}{3a} = \frac{4}{3}, \qquad \tan\theta_2 = \frac{2a}{3a} = \frac{2}{3}.$$

(Indeed $AG = 5a$: a 3–4–5 triangle.)

Hanging from A, then from B

When the wire hangs from A in equilibrium, the only forces on it are the reaction at A and the total weight $3W$ acting at G. For these to balance, their moments about A must cancel, so the weight must pass through A: the line AG is vertical, with G below A. Since BC is parallel to the dashed line in the figure, BC is now tilted at θ_1 to the vertical.

In the same way, when the wire hangs from B, the line BG is vertical, and BC is tilted at θ_2 to the vertical, the other way.

Ask yourself Why is the angle between the two directions of BC equal to $\theta_1 + \theta_2$, rather than $\theta_1 - \theta_2$?

Going from the first position to the second, the wire is turned rigidly, so *every* line in it turns through the same angle. Equivalently, keep the wire fixed on the page and let "downwards" turn instead. In the first position downwards is along GA; in the second it is along GB. So the angle turned is the angle AGB between them. In the figure A and B lie on opposite sides of the dashed line through G, so

$$\angle AGB = \theta_1 + \theta_2.$$

Using the addition formula,

$$\tan(\theta_1 + \theta_2) = \frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1\tan\theta_2} = \frac{\frac{4}{3} + \frac{2}{3}}{1 - \frac{8}{9}} = \frac{2}{\frac{1}{9}} = 18.$$

Both θ_1 and θ_2 are acute, so $0 < \theta_1 + \theta_2 < 180°$, and since its tangent is positive the sum is in fact acute. Hence the angle between the two directions of BC is $\tan^{-1} 18$, as required.

Checks

- Numerically, $\theta_1 = \tan^{-1}\frac{4}{3} \approx 53.13^\circ$ and $\theta_2 = \tan^{-1}\frac{2}{3} \approx 33.69^\circ$, which add to 86.82°, and $\tan^{-1} 18 \approx 86.82^\circ$ too.
- Replacing the wire by 90 000 small equal pieces, averaging their positions to get $G = (4a, 3a)$, and rotating the whole body so that G lies below the support each time, gives the same angle between the two images of BC to nine decimal places.
- Sanity check on G: it lies nearer BC than the missing side AD, as it should, since all the wire on the right is there and none on the left.

Common slips

- Taking G at the centre of the square, $(3a, 3a)$. That would be right for all four sides, but the missing side AD pulls nothing to the left.
- Subtracting the two tilts instead of adding them: BC leans one way when hung from A and the other way when hung from B.

Beyond STEP

The principle "a freely suspended body hangs with its centre of mass below the support" is how you find the centre of mass of an awkward flat shape in practice: hang it from two points and see where the two plumb lines cross. In a first-year course the same idea appears through potential energy: the equilibrium positions are the stationary points of the height of G, and the stable one is the lowest. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats centres of mass and the statics of rigid bodies with many more examples like this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.